	$-\theta$	$\frac{\pi}{2}-\theta$	$\frac{\pi}{2}+\theta$	$\pi-\theta$	$\pi+\theta$	$\frac{3\pi}{2}-\theta$	$\frac{3\pi}{2}+\theta$	$2\pi-\theta$	$2\pi+\theta$
sine	$-\sin\theta$	$\cos\theta$	$\cos\theta$	$\sin\theta$	$-\sin\theta$	$-\cos\theta$	$-\cos\theta$	$-\sin\theta$	$\sin\theta$
cosine	$\cos\theta$	$\sin\theta$	$-\sin\theta$	$-\cos\theta$	$-\cos\theta$	$-\sin\theta$	$\sin\theta$	$\cos\theta$	$\cos\theta$
tangent	$-\tan\theta$	$\cot\theta$	$-\cot\theta$	$-\tan\theta$	$\tan\theta$	$\cot\theta$	$-\cot\theta$	$-\tan\theta$	$\tan\theta$

(i) The corresponding reciprocal ratios can be written using the above table.

(ii) If the allied angles are $\pm\theta, \pi \pm \theta, 2\pi \pm \theta$, that is, angles of the form $2n\frac{\pi}{2} \pm \theta, n \in \mathbb{Z}$, then, the form of trigonometric ratio is unaltered (i.e., sine remains sine, cosine remains cosine etc.,)

(iii) If the allied angles are $\frac{\pi}{2} \pm \theta, \frac{3\pi}{2} \pm \theta$, that is, angles of the form $(2n+1)\frac{\pi}{2} \pm \theta, n \in \mathbb{Z}$ then, the form of trigonometric ratio is altered to its complementary ratio. *i.e.*, it is to add the prefix "co" if it is absent and remove the prefix "co" if it is already present (*i.e.*, sine becomes cosine, cosine become sine etc.,)

(iv) For determining the sign, first find out the quadrant and then attach the appropriate sign (+ or −) according to the quadrant rule "ASTC".

Example 3.11 Find the value of (i) $\sin 150^\circ$ (ii) $\cos 135^\circ$ (iii) $\tan 120^\circ$.

Solution:

(i) $\sin 150^\circ = \sin(90^\circ + 60^\circ) = \cos(60^\circ) = \frac{1}{2}$

(or) $\sin 150^\circ = \sin(180^\circ - 30^\circ) = \sin(30^\circ) = \frac{1}{2}$

(ii) $\cos 135^\circ = \cos(90^\circ + 45^\circ) = -\sin(45^\circ) = -\frac{1}{\sqrt{2}}$

(or) $\cos 135^\circ = \cos(180^\circ - 45^\circ) = -\cos(45^\circ) = -\frac{1}{\sqrt{2}}$

(iii) $\tan 120^\circ = \tan(180^\circ - 60^\circ) = -\tan(60^\circ) = -\sqrt{3}$

(or) write $\tan 120^\circ$ as $\tan(90^\circ + 30^\circ)$ and find the value.

Example 3.12 Find the value of:

(i) $\sin(765^\circ)$ (ii) $\operatorname{cosec}(-1410^\circ)$ (iii) $\cot\left(\frac{-15\pi}{4}\right)$.

Solution:

(i) $\sin 765^\circ = \sin(2 \times 360^\circ + 45^\circ) = \sin 45^\circ = \frac{1}{\sqrt{2}}$

(ii) $\operatorname{cosec}(-1410^\circ) = -\operatorname{cosec}(1410^\circ) = -\operatorname{cosec}(4 \times 360^\circ - 30^\circ) = \operatorname{cosec} 30^\circ = 2$

(iii) $\cot\left(\frac{-15\pi}{4}\right) = -\cot\left(\frac{15\pi}{4}\right) = -\cot\left(4\pi - \frac{\pi}{4}\right) = \cot\frac{\pi}{4} = 1.$

Example 3.13 Prove that $\tan(315^\circ)\cot(-405^\circ) + \cot(495^\circ)\tan(-585^\circ) = 2$

Solution:

L.H.S $= \tan(360^\circ - 45^\circ)[-\cot(360^\circ + 45^\circ)] + \cot(360^\circ + 135^\circ)[-\tan(360^\circ + 225^\circ)]$

$= [-\tan 45^\circ][-\cot 45^\circ] + [-\tan 45^\circ][-\tan 45^\circ]$

$= (-1)(-1) + (-1)(-1) = 2 =$ R.H.S